

Research Conference

Graph Attention Networks for Community Detection:
Leveraging K-Cliques and Feature Similarity

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OVERVIEW

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INTRODUCTION

- **Community detection** is a major field in network science that focuses on hidden groups and their structure in complex networks like citation, social and biological systems.
- This work brings together key ideas from graph similarity measures, centrality-based influence modeling, and Graph Neural Networks, especially attention-based mechanisms.
- Experiments are conducted on **Cora** and **Citeseer** citation networks
- Our approach integrates feature similarity, k-clique structures, Graph Attention Networks, and clustering using Louvain and K-Means.

PROBLEM STATEMENT

- Traditional community detection methods focus only on graph structure, ignoring node attributes.
- This leads to incomplete or less meaningful communities in real-world networks.
- Attributed networks require methods that combine both structure and node features.
- Existing GNN models often treat all neighbors equally, missing key influencers in the network.
- There's a need for an approach that integrates similarity, node influence, and deep learning for better community detection.

METHODOLOGY

Unified Similarity Matrix:

- Combines Jaccard similarity (based on common neighbors) and cosine similarity (based on node features).
- Produces a weighted matrix that captures both structural and attribute-based relationships.
- Formula:
- $S_{ij} = \alpha \cdot \text{jaccard}(i,j) + (1 - \alpha) \cdot \text{cosine}(i,j)$
- Helps identify communities that are both topologically connected and feature-similar.

METHODOLOGY

Clique-Based Attention

- Detects k -cliques, i.e., fully connected subgraphs of size k .
- Nodes in the same clique are considered part of dense local communities.
- Improves attention by focusing on higher order structural patterns, not just immediate neighbors.
- Helps capture closely connected groups often missed by conventional GATs.
- Ensures the model is sensitive to community boundaries within complex graphs.

METHODOLOGY

Centrality Weighting

- Computes eigenvector centrality to assess node importance.
- High centrality indicates a node is connected to other important nodes.
- Integrates centrality into GAT by scaling attention weights:
- $\alpha_{ij}' = \alpha_{ij} \cdot (1 + \beta \cdot \text{sigmoid}(c_i))$
- Influential nodes are prioritized during message passing.
- Makes attention context-aware, not uniformly distributed.

METHODOLOGY

- Scaled GAT uses centrality-modulated attention to learn embeddings that combine neighborhood structure, node influence, and similarity information.
- These embeddings are more robust and better aligned with true community boundaries.
- K-Means clusters nodes in the latent space, while Louvain operates on the similarity graph to maximize modularity.
- Together, they provide flexible, accurate clustering across graph types, with optimal k selected based on modularity, ARI, and silhouette scores.

MODEL ARCHITECTURE

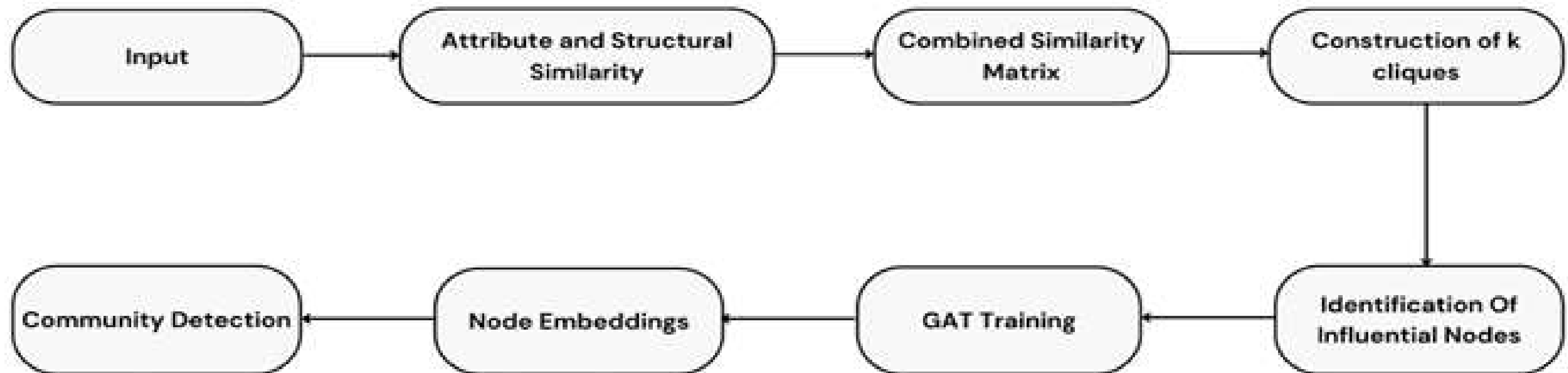


Fig. 1: Block Diagram of the Proposed Methodology

DATASETS

- **Cora** and **Citeseer** citation networks were selected as benchmark datasets due to their widespread use in community detection research and availability of both structural and attribute information.
- Cora contains 2,708 nodes, 5,429 edges, 1,433-dimensional node features, and 7 classes.
- Citeseer includes 3,312 nodes, 4,732 edges, 3,703-dimensional features, and 6 classes.

CORA ANALYSIS

- **Superior Clustering with K-Means:** InfCliqGAT achieved superior clustering with K-Means, recording a modularity score of 0.7617, a silhouette score of 0.2422, NMI of 0.7508, and ARI of 0.7438, along with a test accuracy of 79.52% and F1 score of 77.74%.
- **Louvain Clustering Performance:** Using Louvain clustering, the best results were observed at $k = 4$, with a modularity of 0.6334, silhouette score of 0.1875, NMI of 0.7987, and ARI of 0.8128.
- The values $k = 4$ for Louvain and K-means were chosen after evaluating multiple k values, selecting those that maximized modularity and silhouette scores.

CITeseer Analysis

- **Superior Clustering with K-Means:** InfCliqGAT demonstrated strong clustering performance with K-Means at $k = 5$, achieving a modularity of 0.7494, silhouette score of 0.2383, NMI of 0.7305, and ARI of 0.7696
- **Louvain Clustering Performance:** Using the Louvain algorithm, the best results were obtained at $k = 3$, with a modularity of 0.7357 and silhouette score of 0.1461
- **Optimal k-Clique Selection and Its Impact:** The values $k = 3$ for Louvain and $k=5$ for K-means were chosen after evaluating multiple k values, selecting those that maximized modularity, ARI, and silhouette scores.

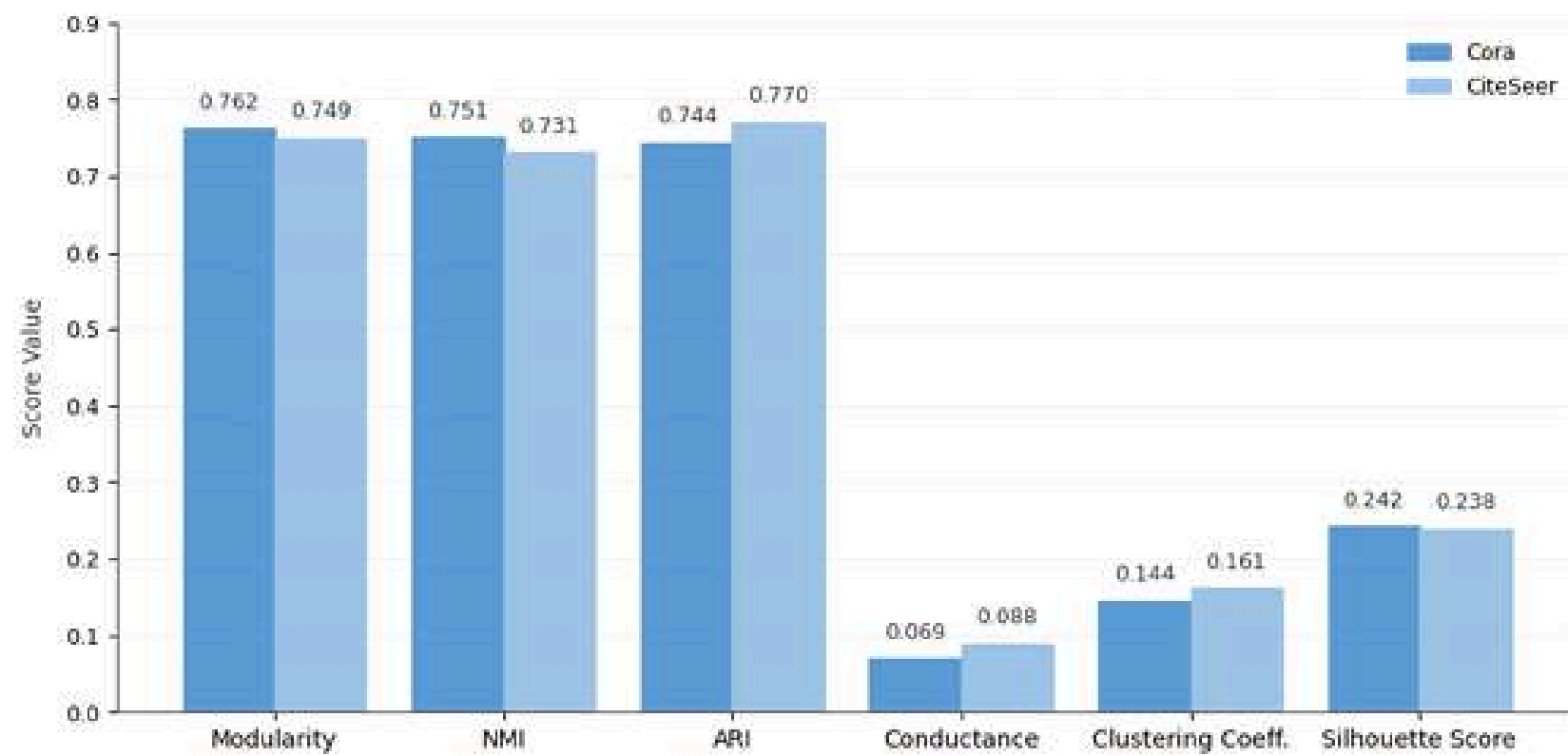


Fig. 2: Evaluation Metrics Comparison for Cora and CiteSeer using K-Means Embeddings

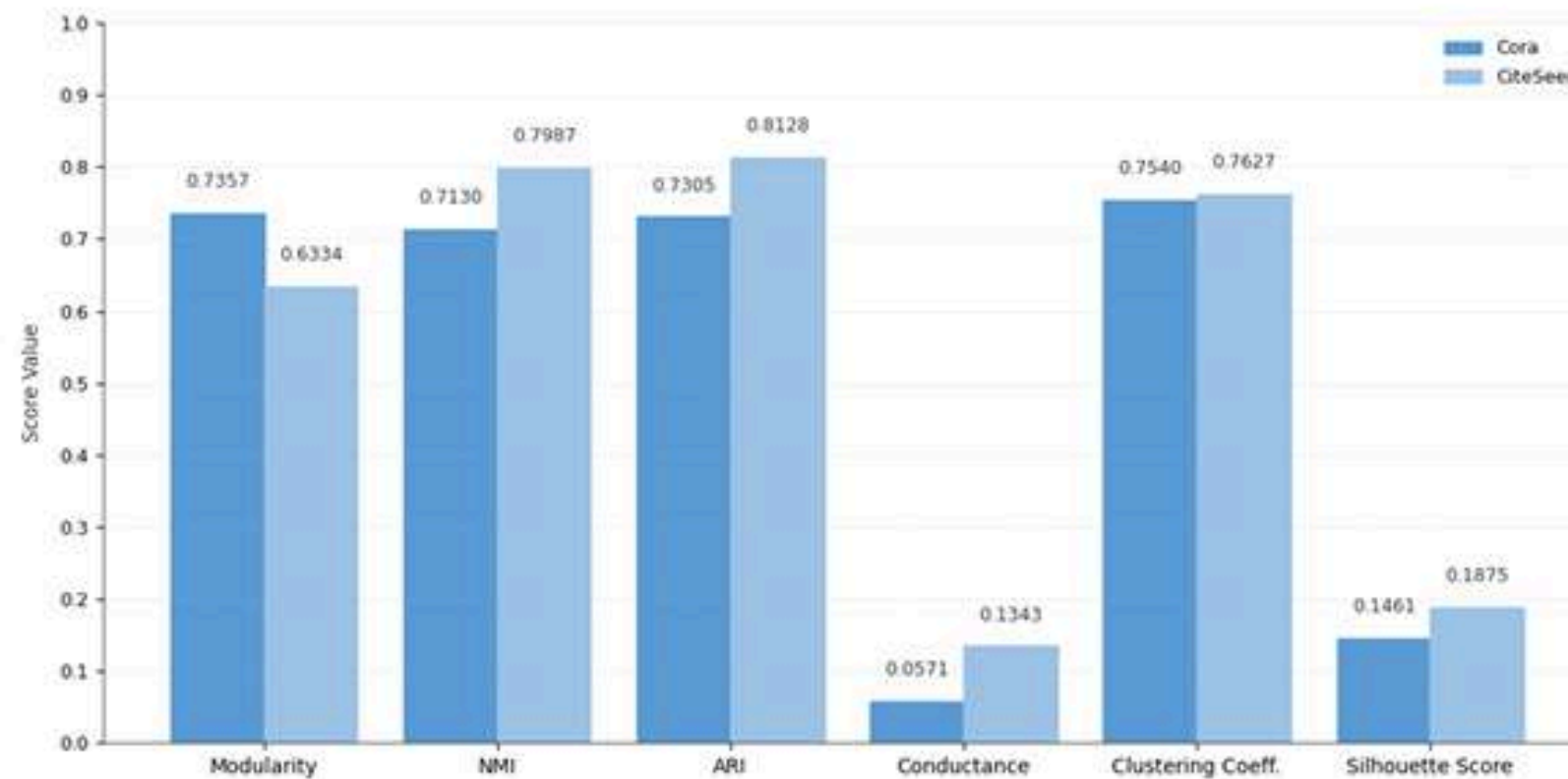


Fig. 3: Evaluation Metrics Comparison for Cora and CiteSeer using the Louvain Algorithm

FINAL RESULTS

- InfCliqGAT demonstrates robust community detection on both Cora and Citeseer, achieving high modularity scores of 0.762 and 0.749 with K-Means clustering. These results indicate more meaningful and well-defined communities compared to traditional baselines.
- By combining structural (Jaccard) and attribute (cosine) similarities, the model achieves strong clustering metrics, with NMI around 0.75 and ARI up to 0.77.
- The selection of optimal k-clique sizes (Cora: 4, Citeseer: 3) and the use of both K-Means and Louvain highlight the model's adaptability in detecting dense, well-separated communities.
- Silhouette scores above 0.23 and low conductance values further validate the compactness and separation of the detected clusters.

COMPARISON WITH EXISTING WORK

Metrics	InfCliqGAT (Citeseer)	GATFELPA (Citeseer)
Modularity Score	0.7494	0.7400
NMI Score	0.7305	0.4820
ARI Score	0.7696	0.4760
Conductance Score	0.0878	0.0800
Clustering Coefficient	0.1611	0.1500
Silhouette Score	0.2383	0.2300
Test Accuracy	0.7360	0.4760
F1 Score (Macro)	0.6912	0.4820

Comparison of Clustering and
Classification Metrics for Citeseer
Dataset

Metrics	InfCliqGAT (Cora)	GATFELPA (Cora)
Modularity Score	0.7617	0.7600
NMI Score	0.7508	0.5630
ARI Score	0.7436	0.5280
Conductance Score	0.0692	0.0700
Clustering Coefficient	0.1443	0.1300
Silhouette Score	0.2422	0.2200
Test Accuracy	0.7952	0.7390
F1 Score (Macro)	0.7774	0.7160

Comparison of Clustering and
Classification Metrics for Cora
Dataset

CONCLUSION

- InfCliqGAT effectively integrates structural and attribute similarities, enhancing community detection in attributed networks.
- The use of k-clique structures and centrality-aware attention improves clustering performance by emphasizing influential nodes.
- Consistent outperformance over baselines like GATFELPA across modularity, NMI, ARI, and silhouette scores demonstrates the model's reliability. Its flexibility with both Louvain and K-Means highlights its adaptability across different clustering strategies.
- The model also achieves strong clustering compactness and separation, supported by high silhouette scores and low conductance values.

FUTURE WORKS

- **Scalability Improvements:** Future work will focus on extending InfCliqGAT to handle large-scale graphs with millions of nodes and edges, enabling real-world deployment in massive network environments.
- **Dynamic and Adaptive Clustering:** Developing adaptive clustering methods for dynamic graphs, where nodes and edges evolve over time will be key to applying InfCliqGAT in real-time scenarios.

Thank you!